# NON-MF GROUPS

SAUERS

We show that not every countable group is MF, where MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. Nonapproximable groups were known for the unnormalized Schatten $p$-norms, $1 \leq p < \infty$; the operator norm case, $p = \infty$, was open and is settled here. The main tool is a one-sided compression criterion that places normal Kazhdan subgroups in the MF radical. Applied to the binary Leavitt group, it shows that the finitely generated simple property-(T) group $H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2))$ has trivial image under every homomorphism to an MF group; consequently its reduced group $C^*$-algebra is a separable stably finite $C^*$-algebra that is not MF. A second application gives a finitely presented torsion-free group with property (T) that has trivial image under every homomorphism to an MF group, and every nontrivial quotient of it has the same property.

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## 1. Introduction

Not every countable group is MF. Our counterexample is

$$H = \mathrm{EL}_{12}(L_{\mathbb{F}_2}(1,2)).$$

The group $H$ is finitely generated, nontrivial, simple, and has property (T), but every homomorphism from $H$ to an MF group is trivial. In particular, $H$ is not MF. Its reduced group $C^*$-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.

More precisely, a countable group $G$ is *MF* if there are positive integers $d_n$ and maps $V_n \colon G \to \mathcal{U}(d_n)$, with $V_n(1) = 1$, such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \qquad (g \in G \setminus \{1\}).$$

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This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

Equivalently, $G$ embeds in the unitary group of the norm matrix corona associated to the sequence $\mathbf{d} = (d_n)$,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \longrightarrow 0\}$$

is the $c_0$-direct sum.

A separable $C^*$-algebra is called MF if it embeds as a $C^*$-subalgebra of a norm matrix corona. For every countable group $G$, the reduced group algebra $C^*_{\mathrm{r}}(G)$ is separable and has a faithful canonical trace, hence is stably finite. A non-MF group therefore gives a separable stably finite reduced group $C^*$-algebra that is not MF (see the proof of Theorem B).

For a countable group $G$, define its MF radical by

$$\mathrm{Rad}_{\mathrm{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

We call a homomorphism $G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ a *corona homomorphism*. Proposition 2.1 shows that $G$ is MF precisely when $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial. If $\mathrm{Rad}_{\mathrm{MF}}(G) = G$, every homomorphism from $G$ to an MF group is trivial.

For a subgroup $L \leq G$, put

$$\mathrm{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression–centralizer defect* by

$$\mathfrak{D}_G(L) = \langle\!\langle [ucu^{-1}, \ell] : u \in \mathrm{Comp}_G(L),\ c \in C_G(L),\ \ell \in L \rangle\!\rangle_G. \tag{1}$$

**Theorem A** (one-sided compression criterion)**.** *Let $G$ be countable and let $L \leq G$ have property* (T)*. If $K \trianglelefteq G$ has property* (T) *and $K \leq \mathfrak{D}_G(L)$, then*

$$K \leq \mathrm{Rad}_{\mathrm{MF}}(G).$$

*In particular, a nontrivial such $K$ makes $G$ non-MF. If $G$ has property* (T) *and $\mathfrak{D}_G(L) = G$, then* $\mathrm{Rad}_{\mathrm{MF}}(G) = G$.

We sketch the argument. Corollary 2.6 gives $\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G)$. That is, every operator norm asymptotic representation of $G$ is asymptotically trivial on $\mathfrak{D}_G(L)$ in normalized Hilbert–Schmidt norm. This holds because conjugation by the compressor moves the Kazhdan projection of $L$ below itself, and stable finiteness of the corona forces equality (Section 2.3). If a corona homomorphism were nontrivial on $K$, the Kazhdan projection of $K$ would cut out a $G$-invariant corner on which $K$ has no nonzero fixed vectors (Lemma 2.7); by the Kazhdan inequality some $s_0 \in K$ would then stay bounded away from the identity in normalized Hilbert–Schmidt norm along a subsequence, contradicting $s_0 \in \mathfrak{D}_G(L)$.

Theorem 2.2 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of $G$ over any field kills $\mathfrak{D}_G(L)$.

Property (T) is not itself an obstruction to MF approximation: every residually finite group is MF, including many Kazhdan groups. In the argument above, property (T) controls the fixed vectors of an exact corona representation. The one-sided compression forces the corresponding Kazhdan projection to commute with the compressor, which makes the Hilbert–Schmidt commutant invariant.

We next apply Theorem A to the group $H$.

Let $L_{\mathbb{F}_2}(1,2)$ denote the binary Leavitt algebra. It is the unital $\mathbb{F}_2$-algebra generated by $s_0, s_1, t_0, t_1$ subject to

$$t_i s_j = \delta_{ij}, \qquad s_0 t_0 + s_1 t_1 = 1. \tag{2}$$

For a unital ring $R$, write $e_{ij}(a) = 1 + aE_{ij}$ and let $\mathrm{EL}_n(R)$ be the subgroup of $\mathrm{GL}_n(R)$ generated by the elements $e_{ij}(a)$, $i \neq j$.

**Theorem B** (the binary Leavitt group)**.** *Put $R = L_{\mathbb{F}_2}(1,2)$ and $H = \mathrm{EL}_{12}(R)$. Then $H$ is finitely generated, nontrivial, simple, has property* (T)*, and*

$$\mathrm{Rad}_{\mathrm{MF}}(H) = H.$$

*Equivalently, every homomorphism from $H$ to an MF group is trivial. In particular, $H$ is non-MF. Its reduced group $C^*$-algebra $C^*_{\mathrm{r}}(H)$ is separable and stably finite, but is not MF.*

Inside $H$, an explicit element $\tau$ satisfies $\tau L \tau^{-1} \leq L$ for the subgroup $L \cong \mathrm{EL}_3(R)$ in the upper left. The element $c = e_{34}(1)$ centralizes $L$, and for $\ell = e_{12}(1)$ the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation $t_1(s_1 t_1)s_1 = 1$ and the Steinberg commutator relations show directly that $d$ normally generates $H$. Since $d \in \mathfrak{D}_H(L)$ and $\mathfrak{D}_H(L)$ is normal, it follows that $\mathfrak{D}_H(L) = H$. Therefore Theorem A gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$.

The second application of Theorem A is to a group without torsion.

**Theorem C** (a torsion-free group with full MF radical)**.** *There is a two-generated, finitely presented, torsion-free, acylindrically hyperbolic group $Q$ with property* (T) *and*

$$\mathrm{Rad}_{\mathrm{MF}}(Q) = Q.$$

*Every nontrivial quotient of $Q$ also equals its own MF radical; in particular, no nontrivial quotient of $Q$ is MF.*

The group $Q$ is a small-cancellation quotient, in the sense of Hull [Hu], of the torsion-free property-(T) group of Fournier-Facio [FF, §2], chosen so that the compression defect normally generates the quotient (Section 4). Thus the obstruction of Theorem A depends neither on torsion, nor on simplicity, nor on the Leavitt algebra.

In a companion paper [Sa] we use $H$, together with an MF permanence theorem for HNN extensions, to show that recognizing MF groups from finite presentations is $\Pi^0_2$-complete. The companion notes [SaN] construct an explicit finitely presented group $E$ from the ascending HNN extension of $\mathbb{Z}^3 \rtimes \mathrm{SL}_3(\mathbb{Z})$ by the doubling endomorphism of $\mathbb{Z}^3$, and prove that $E$ is sofic and not MF; so among finitely presented groups, MF is strictly stronger than sofic and than hyperlinear.

**Relation to prior work.** Blackadar and Kirchberg introduced the notions of MF and NF algebras in a 1994 Oberwolfach abstract [BK94]. They wrote that the classes of NF, strong NF, and separable nuclear stably finite $C^*$-algebras might coincide. Their 1997 paper developed these notions and proved that a separable $C^*$-algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite $C^*$-algebra is quasidiagonal. This was a question about nuclear $C^*$-algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite $C^*$-algebra were MF, then every countable group would be MF because its reduced group $C^*$-algebra is separable and stably finite. Later papers on approximation of groups relate the resulting operator norm problem to Kirchberg's question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite $C^*$-algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group $C^*$-algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that $A *_C A$ is MF whenever $A$ is a separable MF $C^*$-algebra and $C \subseteq A$ is a $C^*$-subalgebra [Sh, Theorem 10]. She also proves that the full group $C^*$-algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten $p$-norms with $1 < p < \infty$ [DGLT, LO]. Bachner–Dogon–Lubotzky later proved nonapproximability for $p = 1$ [BDL]. The operator norm case $p = \infty$ remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]; that is the route taken here.

Leavitt introduced the algebras $L_K(1, n)$ as examples of rings with nonunique module rank [Lev]. The compression used here comes from their defining relations. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of $\mathrm{GL}_n$ over an exchange ring that are normalized by $\mathrm{EL}_n$ [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality $Z(R) = \mathbb{F}_2$, implies that $\mathrm{EL}_{12}(R)$ is simple (Proposition 3.3). Ershov–Jaikin-Zapirain give property (T) for $\mathrm{EL}_n(R)$ whenever $n \geq 3$ and $R$ is a finitely generated unital associative ring [EJZ, Theorem 1.1]. OpenAI used the binary Leavitt unit group to construct the first known nonsofic group [OAI, Chapter 3]. Khanh–Thanh's matrix-ring isomorphisms and their equality $\mathrm{GL}_n(R) = \mathrm{EL}_n(R)$ identify that group with the group $H$ studied here [K–T]. The present argument applies the Leavitt compression to operator norm approximation. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

## 2. One-sided compression and the MF radical

This section proves Theorem A.

**2.1. The MF radical.** The image of a corona homomorphism from $G$ is a countable subgroup of a corona unitary group, hence MF; conversely, every countable MF group embeds in the unitary group of a norm matrix corona. Thus $\mathrm{Rad}_{\mathrm{MF}}(G)$ is also the intersection of the kernels of all homomorphisms from $G$ to MF groups.

**Proposition 2.1** (basic properties of the MF radical)**.** *For every countable group $G$, the subgroup $\mathrm{Rad}_{\mathrm{MF}}(G)$ is fully invariant, the quotient $G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF, and $G$ is MF if and only if $\mathrm{Rad}_{\mathrm{MF}}(G)$ is trivial.*

*Proof.* An intersection of kernels is normal. If $\alpha$ is an endomorphism of $G$, $x \in \mathrm{Rad}_{\mathrm{MF}}(G)$, and $\pi$ is a corona homomorphism from $G$, then $\pi \circ \alpha$ kills $x$. Hence $\pi(\alpha(x)) = 1$ for every $\pi$, which proves full invariance.

Put $R = \mathrm{Rad}_{\mathrm{MF}}(G)$. If $G/R$ is trivial, there is nothing to prove. If $x \in G/R$ is nontrivial and $g \in G$ maps to $x$, then $g \notin R$. Some homomorphism from $G$ to an MF group therefore separates $g$ from the identity. It kills $R$ and descends to a homomorphism from $G/R$ that separates $x$. Thus $G/R$ is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

If $G$ is MF, a corona embedding of $G$ has trivial kernel, so $\mathrm{Rad}_{\mathrm{MF}}(G) = 1$. Conversely, if $\mathrm{Rad}_{\mathrm{MF}}(G) = 1$, then $G = G/\mathrm{Rad}_{\mathrm{MF}}(G)$ is MF. ☐

**2.2. Finite dimension.**

**Theorem 2.2** (finite-dimensional commutant rigidity)**.** *Let $G$ be a group, let $L \leq G$, and suppose $u \in G$ satisfies $uLu^{-1} \leq L$. If $k$ is a field, $V$ is a finite-dimensional $k$-vector space, and $\rho\colon G \to \mathrm{GL}(V)$ is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

*where $\rho(L)'$ is the commutant of $\rho(L)$ in $\mathrm{End}_k(V)$. Consequently, if $c \in C_G(L)$, then*

$$\rho([ucu^{-1}, \ell]) = 1 \qquad (\ell \in L).$$

*Proof.* Put $C = \rho(L)'$. If $x \in C$, $h \in L$, and $h' = uhu^{-1} \in L$, then

$$\begin{aligned}\rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u)\\ &= \rho(u)^{-1}x\rho(h')\rho(u)\\ &= \rho(u)^{-1}x\rho(u)\rho(h).\end{aligned}$$

Thus $\rho(u)^{-1}C\rho(u) \subseteq C$. Conjugation by $\rho(u)^{-1}$ is injective, and $C$ is finite-dimensional. Its restriction to $C$ is therefore surjective, so the inclusion is equality. Conjugating by $\rho(u)$ gives $\rho(u)C\rho(u)^{-1} = C$.

If $c \in C_G(L)$, then $\rho(c) \in C$, hence $\rho(ucu^{-1}) \in C$. Therefore $\rho([ucu^{-1}, \ell]) = 1$ for every $\ell \in L$. $\square$

The finite-dimensionality hypothesis cannot be dropped: an injective linear self-map of an infinite-dimensional commutant need not be surjective.

2.3. **Kazhdan transport in normalized Hilbert–Schmidt norm.** For $a \in M_d(\mathbb{C})$ write

$$\|a\|_2 = \mathrm{tr}_d(a^*a)^{1/2}, \qquad \mathrm{tr}_d = \frac{1}{d}\,\mathrm{Tr}\,.$$

An *operator norm asymptotic representation* of a countable group $G$ is a sequence of maps $V_n \colon G \to \mathcal{U}(d_n)$ such that $V_n(1) = 1$ and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \qquad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so $K_2(V)$ is a normal subgroup of $G$. Define

$$R_{\infty\to 2}(G) = \bigcap_V K_2(V), \tag{3}$$

where $V$ ranges over all operator norm asymptotic representations of $G$.

For $L \leq G$, let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \;\middle|\; \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of $V(L)$. Coordinatewise conjugation by $V_n(g)$ defines a bijection, denoted $\mathrm{Ad}(V(g))$, of the bounded matrix sequences.

**Lemma 2.3.** *For every sequence of positive integers $(m_n)$, the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

*is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.*

*Proof.* Suppose that $v^*v = 1$ in the corona, and let $(x_n)$ be a bounded lift of $v$. Then $x_n^*x_n \to 1$ in norm. For all sufficiently large $n$, the matrix $x_n$ is invertible, and the unitary $x_n(x_n^*x_n)^{-1/2}$ differs from $x_n$ by $o(1)$. Thus $v$ is represented by unitaries and $vv^* = 1$. The same argument applies to every matrix amplification of the corona. If $A$ is a stably finite unital $C^*$-algebra, equivalent projections $p \leq q$ satisfy $p = q$. Indeed, if $w^*w = q$ and $ww^* = p$, then $w \in qAq$ is an isometry in the corner. The corner is finite because $w + (1-q)$ is an isometry in $A$. Consequently $ww^* = q$, and hence $p = q$. □

**Lemma 2.4** (one-sided order for the Kazhdan projection)**.** *Let $L$ have property* (T)*, let $B$ be a unital $C^*$-algebra, and let $\pi\colon L \to \mathcal{U}(B)$ be a homomorphism. Denote by $P \in B$ the image of the Kazhdan projection under the extension $C^*_{\max}(L) \to B$. If $U \in \mathcal{U}(B)$ satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

*then*

$$U^*PU \leq P.$$

*Proof.* Represent $B$ faithfully and nondegenerately on a Hilbert space $\mathcal{H}$. The represented projection $P$ is the orthogonal projection onto $\operatorname{Fix}\pi(L)$. If $\xi$ is $L$-fixed and $\ell \in L$, then

$$\pi(\ell)U^*\xi = U^*\big(U\pi(\ell)U^*\big)\xi = U^*\xi,$$

because $U\pi(\ell)U^*$ belongs to $\pi(L)$. Hence $U^* \operatorname{Fix}\pi(L) \subseteq \operatorname{Fix}\pi(L)$, so the range projection $U^*PU$ is dominated by $P$ in $B(\mathcal{H})$. Faithfulness of the representation gives the same projection inequality in $B$. □

**Theorem 2.5** (one-sided Kazhdan transport)**.** *Let $L \leq G$ have property* (T)*, and let $u \in G$ satisfy $uLu^{-1} \leq L$. Let $(V_n)$ be an operator norm asymptotic representation of $G$. Then*

$$\operatorname{Ad}(V(u))\big(\mathcal{C}_2(V, L)\big) = \mathcal{C}_2(V, L).$$

*Proof.* Fix $x = (x_n) \in \mathcal{C}_2(V, L)$ and let $\omega$ be a free ultrafilter. Regard $M_{d_n}(\mathbb{C})$ as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces $\mathcal{K}_\omega$. Conjugation by the $V_n(g)$ defines a unitary representation $\sigma$ of $G$ on $\mathcal{K}_\omega$. Indeed, for unitaries $A, B$ one has

$$\|\operatorname{Ad}(A) - \operatorname{Ad}(B)\| \leq 2\,\|A - B\|\,,$$

so the conjugation actions are multiplicative modulo $c_0$ because the multiplicative defects of $(V_n)$ tend to zero in operator norm. The class $\xi = [x_n]_\omega$ is fixed by $\sigma(L)$.

For $g \in G$, let $\widetilde{\sigma}(g)$ be the class of the coordinate operators $\operatorname{Ad}(V_n(g))$ in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each $M_{d_n}(\mathbb{C})$, this is a norm matrix corona with coordinate sizes $d_n^2$. By the preceding estimate, $g \mapsto \widetilde{\sigma}(g)$ is a homomorphism $G \to \mathcal{U}(\mathcal{B})$. Its restriction to $L$ therefore extends to a $*$-homomorphism $C^*_{\max}(L) \to \mathcal{B}$. Let $P \in \mathcal{B}$ be the image of the Kazhdan projection [AW]. Under the natural representation of $\mathcal{B}$ on $\mathcal{K}_\omega$, its range is $\operatorname{Fix}\sigma(L)$. The projection identity below is proved inside $\mathcal{B}$; the representation of $\mathcal{B}$ on $\mathcal{K}_\omega$ is used only afterwards. Since $uLu^{-1} \leq L$, the unitary $U = \widetilde{\sigma}(u)$ satisfies the hypothesis of Lemma 2.4, and hence $U^*PU \le P$ inside $\mathcal{B}$. The two projections are unitarily equivalent. Lemma 2.3 therefore gives $U^*PU = P$, so $U$ commutes with $P$. Hence both $U\xi$ and $U^*\xi$ are fixed by $\sigma(L)$. Since $\omega$ was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus $\mathrm{Ad}(V(u))$ and its inverse both preserve $\mathcal{C}_2(V, L)$, which proves the equality. □

**Corollary 2.6.** *Let $L \le G$ have property* (T)*, let $uLu^{-1} \le L$, and let $c \in C_G(L)$. Then*

$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G) \qquad (\ell \in L).$$

*Proof.* Let $(V_n)$ be an operator norm asymptotic representation. If $c \in C_G(L)$, then $(V_n(c)) \in \mathcal{C}_2(V, L)$. By Theorem 2.5, the same is true of $(V_n(u)V_n(c)V_n(u)^*)$. Asymptotic multiplicativity identifies this sequence in operator norm with $(V_n(ucu^{-1}))$; explicitly,

$$\Big\| V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1}) \Big\| \longrightarrow 0.$$

Therefore

$$\Big\| V_n([ucu^{-1}, \ell]) - 1 \Big\|_2 \longrightarrow 0 \qquad (\ell \in L).$$

Intersecting over $(V_n)$ proves the claim. □

2.4. **Normal Kazhdan subgroups and the MF radical.** Let $K \trianglelefteq G$ have property (T), and let $p$ be the image of its Kazhdan projection under a corona homomorphism of $G$. Normality of $K$ implies that $p$ commutes with the image of $G$. Hence $q = 1 - p$ also commutes with that image, and in every nondegenerate representation of the corner $q\mathcal{Q}_{\mathbf{d}}q$ the induced representation of $K$ has no nonzero fixed vectors.

**Lemma 2.7** (central corona corners)**.** *Let $\rho\colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ be a homomorphism from a countable group, and let $q \in \mathcal{Q}_{\mathbf{d}}$ be a nonzero projection commuting with $\rho(G)$. Then, after passing to an infinite coordinate subsequence, there are nonzero projections $q_n \in M_{d_n}(\mathbb{C})$, integers $r_n = \mathrm{rank}(q_n)$, unitary identifications $J_n\colon \mathbb{C}^{r_n} \to q_n\mathbb{C}^{d_n}$, and an operator norm asymptotic representation*

$$W_n\colon G \longrightarrow \mathcal{U}(r_n)$$

*with $W_n(1) = I_{r_n}$. In the corona over the retained coordinates, the class of $(J_nW_n(g)J_n^*)$ is the coordinate restriction of $q\rho(g)$.*

*Proof.* Lift $q$ to projections $q_n$, and lift each $\rho(g)$ to unitaries $U_n(g)$, choosing $U_n(1) = I_{d_n}$. The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_nU_n(g) - U_n(g)q_n\| \longrightarrow 0 \qquad (g \in G).$$

Consequently $q_nU_n(g)q_n$, viewed on $q_n\mathbb{C}^{d_n}$, has both unitarity defects converging to zero. For each fixed $g$, let $\widetilde{W}_n(g)$ be its polar correction whenever both defects are at most $1/2$, and set $\widetilde{W}_n(g) = q_n$ at the finitely many earlier stages. Then $\widetilde{W}_n(g)$ is unitary in the corner and satisfies

$$\left\| \widetilde{W}_n(g) - q_n U_n(g) q_n \right\| \longrightarrow 0.$$

For $g = 1$, the compression is already the identity $q_n$ of the corner, so take $\widetilde{W}_n(1) = q_n$. Since $q \neq 0$, infinitely many $q_n$ are nonzero. Retain and relabel those coordinates. Put $r_n = \operatorname{rank}(q_n)$, choose a unitary $J_n \colon \mathbb{C}^{r_n} \to q_n \mathbb{C}^{d_n}$, and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by $J_n$ preserves operator norms and multiplicative defects. Thus $(W_n)$ is an operator norm asymptotic representation with $W_n(1) = I_{r_n}$. The displayed estimate identifies the class of $(J_n W_n(g) J_n^*)$ with the coordinate restriction of $q\rho(g)$. □

**Theorem 2.8** (normal Kazhdan radical theorem)**.** *Let $G$ be countable and let $K \trianglelefteq G$ have property* (T) *with $K \leq R_{\infty\to 2}(G)$. Then*

$$K \leq \operatorname{Rad}_{\mathrm{MF}}(G).$$

*Proof.* Suppose that a homomorphism $\Theta \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ is nontrivial on $K$. Replace $G$ by $\Theta(G)$, replace $K$ by $\Theta(K)$, and denote the resulting inclusion into the corona again by $\Theta$. The new $K$ is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. It is contained in $R_{\infty\to 2}(\Theta(G))$, because an operator norm asymptotic representation of $\Theta(G)$, precomposed with the quotient map, is one of the original group.

Let $p_K \in C^*_{\max}(K)$ be the Kazhdan projection and let $p$ be its image in $\mathcal{Q}_{\mathbf{d}}$. Under any faithful representation of the corona, $p$ is the projection onto the $K$-fixed vectors. Normality of $K$ gives

$$\Theta(g) p \Theta(g)^* = p \qquad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges $\Theta(g) \operatorname{Fix} \Theta(K)$ and $\operatorname{Fix} \Theta(K)$; these agree because $gKg^{-1} = K$. Put $q = 1 - p$. The projection $q$ is nonzero: if $p = 1$, then every vector is fixed by $K$, so every element of $K$ has trivial corona image. Lemma 2.7 gives positive integers $r_n$ and an operator norm asymptotic representation $W_n \colon G \to \mathcal{U}(r_n)$. Let $\widehat{\Theta} \colon G \to \mathcal{U}(\mathcal{Q}_{\mathbf{r}})$ be its corona homomorphism. Under the coordinatewise corner identifications, $\widehat{\Theta}$ is the coordinate restriction of $g \mapsto q\Theta(g)$.

Choose a finite Kazhdan set $S \subseteq K$ and a Kazhdan constant $\kappa > 0$. Represent $\mathcal{Q}_{\mathbf{r}}$ faithfully. The Kazhdan projection of $K$ under $\widehat{\Theta}$ is zero, so $\widehat{\Theta}|_K$ has no nonzero fixed vectors, and the Kazhdan inequality gives, for the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\widehat{\Theta}(s) - 1)^* (\widehat{\Theta}(s) - 1)$$

the inequality
$$b \geq \frac{\kappa^2}{|S|}\, 1.$$

The coordinate elements
$$b_n = \frac{1}{|S|}\sum_{s\in S}(W_n(s) - I_{r_n})^*(W_n(s) - I_{r_n})$$
represent $b$. Taking the normalized trace on $M_{r_n}(\mathbb{C})$ yields
$$\frac{1}{|S|}\sum_{s\in S}\|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$
The order inequality for $b$ says that the negative part of $b_n - (\kappa^2/|S|)I_{r_n}$ converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed $s_0 \in S$ therefore stays a positive Hilbert–Schmidt distance from $I_{r_n}$. This contradicts $s_0 \in K \leq R_{\infty\to 2}(G)$, established after passing to $\Theta(G)$. Thus every corona homomorphism kills $K$, as required. □

*Proof of Theorem A.* For every $u \in \mathrm{Comp}_G(L)$, $c \in C_G(L)$, and $\ell \in L$, Corollary 2.6 gives
$$[ucu^{-1}, \ell] \in R_{\infty\to 2}(G).$$
Normality of $R_{\infty\to 2}(G)$ therefore gives
$$\mathfrak{D}_G(L) \leq R_{\infty\to 2}(G).$$
Since $K \leq \mathfrak{D}_G(L)$, Theorem 2.8 now gives $K \leq \mathrm{Rad}_{\mathrm{MF}}(G)$. Taking a nontrivial $K$ gives the non-MF conclusion. Taking $K = G = \mathfrak{D}_G(L)$ gives $\mathrm{Rad}_{\mathrm{MF}}(G) = G$. □

By Theorem 2.2, every finite-dimensional linear representation of $G$ kills the generators of $\mathfrak{D}_G(L)$, and its kernel is normal, so it kills $\mathfrak{D}_G(L)$. Consequently, $\mathfrak{D}_G(L) = 1$ if $G$ has a faithful finite-dimensional linear representation. The same conclusion holds if $G$ is residually finite, because every nonidentity element then survives in a finite permutation representation. If $G$ is amenable, its property-(T) subgroup $L$ is finite [BHV]; then $uLu^{-1} = L$, so $ucu^{-1}$ centralizes $L$ for every $u \in \mathrm{Comp}_G(L)$, and again $\mathfrak{D}_G(L) = 1$.

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.5.

## 3. The binary Leavitt group

This section proves Theorem B. The relation $s_0t_0 + s_1t_1 = 1$ of the binary Leavitt algebra gives an element $\tau \in H$ with $\tau L\tau^{-1} \leq L$ for a copy $L$ of $\mathrm{EL}_3(R)$, and a commutator $d \neq 1$ in $\mathfrak{D}_H(L)$; since $d$ normally generates $H$, $\mathfrak{D}_H(L) = H$, and Theorem A applies.

3.1. **The self-compression.** Throughout this section $R = L_{\mathbb{F}_2}(1,2)$ and

$$p = s_0 t_0, \qquad q = s_1 t_1.$$

The relations (2) give

$$p + q = 1, \qquad t_1 q s_1 = 1. \tag{4}$$

In particular $q \neq 0$.

We use indices $0, \dots, 11$ for $12 \times 12$ matrices. We identify $\mathrm{GL}_3(R)$ with the subgroup of $\mathrm{GL}_{12}(R)$ consisting of the matrices $\mathrm{diag}(A, I_9)$. Define a map $\Psi\colon M_3(R) \to M_3(R)$ by

$$\Psi(A) = qI_3 + s_0 A t_0. \tag{5}$$

Here $s_0 A t_0$ denotes the matrix $(s_0 a_{ij} t_0)_{ij}$. The relations $q^2 = q$, $qs_0 = t_0 q = 0$, and $t_0 s_0 = 1$, all consequences of (2), show that $\Psi$ is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0 I_3 t_0 = qI_3 + pI_3 = (p+q)I_3 = I_3,$$

so $\Psi$ preserves the identity. Finally, $t_0 \Psi(A) s_0 = A$, so $\Psi$ is injective. Hence $\Psi$ restricts to an injective group homomorphism $\mathrm{GL}_3(R) \to \mathrm{GL}_3(R)$.

To realize $\Psi$ by conjugation on the embedded copy of $\mathrm{GL}_3(R)$, define the following $6 \times 6$ block matrices:

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \qquad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives $XY = YX = I_6$, so $Y = X^{-1}$. Put

$$\tau = \mathrm{diag}(X, Y) \in \mathrm{GL}_{12}(R). \tag{6}$$

A direct block multiplication gives

$$\tau \, \mathrm{diag}(A, I_9) \, \tau^{-1} = \mathrm{diag}(\Psi(A), I_9). \tag{7}$$

**Lemma 3.1.** *The matrix $\tau$ defined in (6) belongs to $\mathrm{EL}_{12}(R)$.*

*Proof.* Put $I = I_6$. The Whitehead factorization gives

$$\mathrm{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}. \tag{8}$$

If $B = (b_{rs})_{r,s=0}^5 \in M_6(R)$, then

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{r,6+s}(b_{rs}), \qquad \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \prod_{r,s=0}^{5} e_{6+r,s}(b_{rs}).$$

In either product, distinct off-diagonal matrix units have product zero, so no cross-terms occur. Every factor on the right-hand side of (8) therefore belongs to $\mathrm{EL}_{12}(R)$. Since $Y = X^{-1}$, its left-hand side is $\tau$, and the result follows. $\square$

Set $H = \mathrm{EL}_{12}(R)$, and let $L = \mathrm{EL}_3(R) \leq H$ denote the image of the embedding $A \mapsto \mathrm{diag}(A, I_9)$.

**Proposition 3.2.** *The groups $H$ and $L$ have property* (T)*, and $\tau \in H$ satisfies*

$$\tau L \tau^{-1} \leq L. \tag{9}$$

*Proof.* The ring $R$ is generated by $s_0, s_1, t_0, t_1$ as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for $\mathrm{EL}_{12}(R)$ and $\mathrm{EL}_3(R)$ [EJZ, Theorem 1.1]. Lemma 3.1 gives $\tau \in H$.

For $i \neq j$ and $a \in R$, equations (7) and (5) give

$$\tau e_{ij}(a)\tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of $L$. These matrices generate $L$, so (9) follows. □

3.2. **Simplicity and the full MF radical.**

**Proposition 3.3.** *The group $H = \mathrm{EL}_{12}(R)$ is nontrivial and simple.*

*Proof.* The ring $R$ is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let $N \trianglelefteq H$. Since $H = \mathrm{EL}_{12}(R)$, the subgroup $N \leq \mathrm{GL}_{12}(R)$ is normalized by $\mathrm{EL}_{12}(R)$. Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal $I \trianglelefteq R$ such that

$$\mathrm{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of $R$ gives $I = 0$ or $I = R$. If $I = R$, then $\mathrm{EL}_{12}(R, I) = H$, so $N = H$.

Suppose that $I = 0$. Then $N \leq C_{12}(R, 0) = Z(\mathrm{GL}_{12}(R))$. A matrix that commutes with every $e_{ij}(1)$ is a scalar matrix $\lambda I_{12}$. Commutation with $e_{ij}(a)$ for arbitrary $a \in R$ then gives $\lambda a = a\lambda$, so $\lambda \in Z(R)^{\times}$. Since $Z(R) = \mathbb{F}_2$ [AC, Corollary 4.3], we have $\lambda = 1$ and $N = 1$. Finally, $e_{01}(1) \neq 1$, so $H$ is nontrivial. □

**Proposition 3.4.** *Let*

$$c = e_{34}(1), \qquad \ell = e_{12}(1), \qquad d = [\tau c \tau^{-1}, \ell]$$

*in $H$. Then*

$$c \in C_H(L), \qquad d = e_{02}(q) \neq 1, \qquad \langle\!\langle d \rangle\!\rangle_H = H.$$

*Proof.* For $i, j \in \{0, 1, 2\}$ with $i \neq j$ and $a \in R$, we have $j \neq 3$ and $i \neq 4$. Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate $L$, so $c = e_{34}(1)$ belongs to $C_H(L)$.

Direct multiplication of the matrices in (6) gives

$$\tau c \tau^{-1} = e_{01}(q) e_{34}(1). \tag{10}$$

The second factor commutes with $\ell = e_{12}(1)$. The identity $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$ therefore gives $d = e_{02}(q)$, which is nontrivial because $q \neq 0$.

Let $N = \langle\langle d \rangle\rangle_H$. Since $t_1 q s_1 = 1$, the Steinberg relations give

$$[d, e_{23}(s_1)] = e_{03}(qs_1) \in N, \qquad [e_{10}(t_1), e_{03}(qs_1)] = e_{13}(1) \in N.$$

Conjugation by elementary Weyl matrices gives $e_{ij}(1) \in N$ for every $i \neq j$. For $a \in R$ and distinct $i, j, k$,

$$[e_{ik}(a), e_{kj}(1)] = e_{ij}(a) \in N.$$

Thus $N$ contains every elementary generator of $H$, so $N = H$. □

*Proof of Theorem B.* Let $Y$ be the set of elements $e_{ij}(a)$ with $i \neq j$ and $a \in \{1, s_0, s_1, t_0, t_1\}$. Since $e_{ij}(a+b) = e_{ij}(a)e_{ij}(b)$ and $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$ for distinct $i, j, k$, and since every element of $R$ is a sum of monomials in $s_0, s_1, t_0, t_1$, induction on the length of a monomial shows that every $e_{ij}(a)$ lies in $\langle Y \rangle$; there is always a third index because $12 \geq 3$. Thus $H = \langle Y \rangle$ is finitely generated, and $H$ has property (T) by the Ershov–Jaikin-Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (9), the centrality of $c$ relative to $L$, and Proposition 3.4 show that $d \in \mathfrak{D}_H(L)$. The subgroup $\mathfrak{D}_H(L)$ is normal and Proposition 3.4 gives $\langle\langle d \rangle\rangle_H = H$, so $\mathfrak{D}_H(L) = H$. The group $H$ has property (T), so Theorem A, applied with $K = H$, gives $\mathrm{Rad}_{\mathrm{MF}}(H) = H$. Proposition 3.3 shows that $H$ is nontrivial and simple.

If $M$ is MF and $f\colon H \to M$ is a homomorphism, compose $f$ with a faithful corona embedding of $M$. Since $\mathrm{Rad}_{\mathrm{MF}}(H) = H$, this composition and hence $f$ are trivial. If $H$ were MF, its identity homomorphism would contradict this conclusion.

Finally, $C^*_{\mathrm{r}}(H)$ is separable because $H$ is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding $e\colon C^*_{\mathrm{r}}(H) \to \mathcal{Q}_{\mathbf{d}}$ existed and $p = e(1)$, then $h \mapsto e(\lambda_h) + (1-p)$ would embed $H$ in $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$. This contradicts the fact that $H$ is not MF. □

## 4. A torsion-free group with full MF radical

A group is acylindrically hyperbolic if it admits a non-elementary acylindrical action on a hyperbolic space [Os]. For such a group $G$, Hull [Hu, Theorem 3.12] provides a generating set $A$ such that the Cayley graph $\Gamma(G, A)$ is hyperbolic and the action of $G$ on it is acylindrical and non-elementary; a subgroup $N \leq G$ is *suitable* with respect to $A$ if $N$ acts non-elementarily on $\Gamma(G, A)$ and normalizes no nontrivial finite subgroup of $G$ [Hu, Definition 1.4]. We use Hull's small cancellation theorem [Hu, Theorem 7.1] in the following form.

**Theorem 4.1** (Hull's small cancellation theorem)**.** *Let $G$ be acylindrically hyperbolic, let $N \leq G$ be suitable with respect to $A$, and let $t_1, \ldots, t_m \in G$ and a finite subset $F \subseteq G$ be given. Then there is a surjective homomorphism $q\colon G \to Q$ such that $Q$ is acylindrically hyperbolic, $q|_F$ is injective, $q(t_i) \in q(N)$ for all $i$, and every element of finite order in $Q$ is the image of an element of the same order in $G$.*

Moreover the kernel of $q$ is the normal closure of $m$ elements; this is read off from Hull's proof: the case $m = 1$ adds the single relator $W = t_1^{-1} h_1^{m_1} h_2^{l_1} \cdots h_1^{m_n} h_2^{l_n}$ for suitable $h_1, h_2 \in N$, and the general case is an induction on $m$.

**Lemma 4.2** (saturation)**.** *Let $G$ be finitely presented, torsion-free, and acylindrically hyperbolic, let $N \trianglelefteq G$ be nontrivial, and let $F \subseteq G$ be finite. Then there is a surjective homomorphism $q\colon G \to Q$ such that $Q$ is two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q|_F$ is injective, and $q(N) = Q$.*

*Proof.* Fix $A$ as above. Since $G$ is torsion-free and $N \neq 1$, $N$ is infinite, so $|N \cap gNg^{-1}| = \infty$ for every $g \in G$, and by Osin [Os, Lemma 7.1] $N$ acts non-elementarily on $\Gamma(G, A)$. The only finite subgroup of $G$ is trivial, so $N$ is suitable. By Hull [Hu, Corollary 5.7 and Lemma 5.8], $N$ contains two elements $h_1, h_2$ such that $N_0 = \langle h_1, h_2 \rangle$ is again suitable. Apply Theorem 4.1 to $N_0$ with $t_1, \ldots, t_m$ a finite generating set of $G$ and with the set $F$. Then

$$Q = \langle q(t_1), \ldots, q(t_m) \rangle \leq q(N_0) \leq q(N) \leq Q,$$

so $Q = q(N_0) = \langle q(h_1), q(h_2) \rangle$ and $q(N) = Q$. The group $Q$ is torsion-free because $G$ is, and finitely presented because it is $G$ with $m$ relators added. ☐

We now recall the construction of Fournier-Facio [FF, §2]; the references for the ingredients are given there. Let $U$ be a universal finitely presented torsion-free group, one containing a copy of every finitely presented torsion-free group; let $S$ be a finitely presented infinite simple torsion-free group; and let $H_0$ be a torsion-free hyperbolic group with property (T). Small cancellation over the relatively hyperbolic pair $(U * H_0, U)$ embeds $U$ in a finitely presented torsion-free quotient $P$ of $H_0$, which therefore has property (T). By universality, $P$ contains a subgroup $P_1 \times P_2 \times S$ with $P_i \cong P$. Let $E$ be the double HNN extension of $P$ with stable letters $u_1, u_2$ and $u_i P u_i^{-1} = P_i$; it is finitely presented, torsion-free, and acylindrically hyperbolic. Hull's common quotient theorem [Hu, Corollary 7.4] applied to $E$ and $H_0$ gives a surjection $\pi\colon E \to G_0$ onto a finitely presented torsion-free group with property (T), injective on a prescribed finite subset of $E$; since $E$ is finitely generated, the proof of that corollary produces $G_0$ acylindrically hyperbolic. We run the construction with the prescribed finite subset containing an element $s \neq 1$ of $S$. Then $\pi(S) \neq 1$, so $\pi|_S$ is injective because $S$ is simple.

Put $\Gamma = \pi(P)$, $t = \pi(u_1)$, and $J = t^{-1}\pi(S)t$. Since $S$ commutes with $P_1 = u_1 P u_1^{-1}$, the subgroup $J$ centralizes $\Gamma$, and $t\Gamma t^{-1} = \pi(P_1) \leq \Gamma$, so $t \in \mathrm{Comp}_{G_0}(\Gamma)$. Moreover $tJt^{-1} = \pi(S) \leq \Gamma$, and $\Gamma$ has property (T) as a quotient of $P$.

**Lemma 4.3.** *Let $\rho\colon G_0 \to L$ be a surjective homomorphism with $\rho(\pi(S)) \neq 1$. Then $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$.*

*Proof.* Write $\Gamma_L = \rho(\Gamma)$, $t_L = \rho(t)$, and $J_L = \rho(J)$. Then $t_L \Gamma_L t_L^{-1} \leq \Gamma_L$, $J_L$ centralizes $\Gamma_L$, and $t_L J_L t_L^{-1} = \rho(\pi(S)) \leq \Gamma_L$. For $c \in J_L$ and $\ell \in \Gamma_L$, the commutator $[t_L c t_L^{-1}, \ell]$ lies in $\mathfrak{D}_L(\Gamma_L)$ by (1). As $c$ ranges over $J_L$, the element $t_L c t_L^{-1}$ ranges over $\rho(\pi(S))$, and $\ell$ may be taken in $\rho(\pi(S)) \leq \Gamma_L$; hence the commutator subgroup of $\rho(\pi(S))$ lies in $\mathfrak{D}_L(\Gamma_L)$. Since $\rho(\pi(S)) \neq 1$ and $S$ is simple, $\rho(\pi(S)) \cong S$ is perfect, so $\rho(\pi(S)) \leq \mathfrak{D}_L(\Gamma_L)$. □

*Proof of Theorem C.* Let $N$ be the normal closure of $\pi(S)$ in $G_0$. It is nontrivial, so Lemma 4.2 applied to $G_0$, $N$, and $F = \{1, \pi(s)\}$ gives $q\colon G_0 \to Q$ with $Q$ two-generated, finitely presented, torsion-free, and acylindrically hyperbolic, $q(N) = Q$, and $q(\pi(s)) \neq 1$. The group $Q$ has property (T) as a quotient of $G_0$.

Let $r\colon Q \to L$ be a surjection onto a nontrivial group and put $\rho = r \circ q$. Since $q(N) = Q$, the normal closure of $\rho(\pi(S))$ in $L$ is $\rho(N) = L$; as $L \neq 1$, this forces $\rho(\pi(S)) \neq 1$. By Lemma 4.3, $\rho(\pi(S)) \leq \mathfrak{D}_L(\rho(\Gamma))$, and since $\mathfrak{D}_L(\rho(\Gamma))$ is normal in $L$ and contains a normal generating set of $L$, it equals $L$. The group $L$ has property (T) as a quotient of $Q$, and $\rho(\Gamma)$ has property (T) as a quotient of $\Gamma$. Theorem A, applied to $L$ with the subgroup $\rho(\Gamma)$ and $K = L$, gives $\mathrm{Rad}_{\mathrm{MF}}(L) = L$. Taking $r$ the identity gives $\mathrm{Rad}_{\mathrm{MF}}(Q) = Q$, and a nontrivial group equal to its own MF radical is not MF. □


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